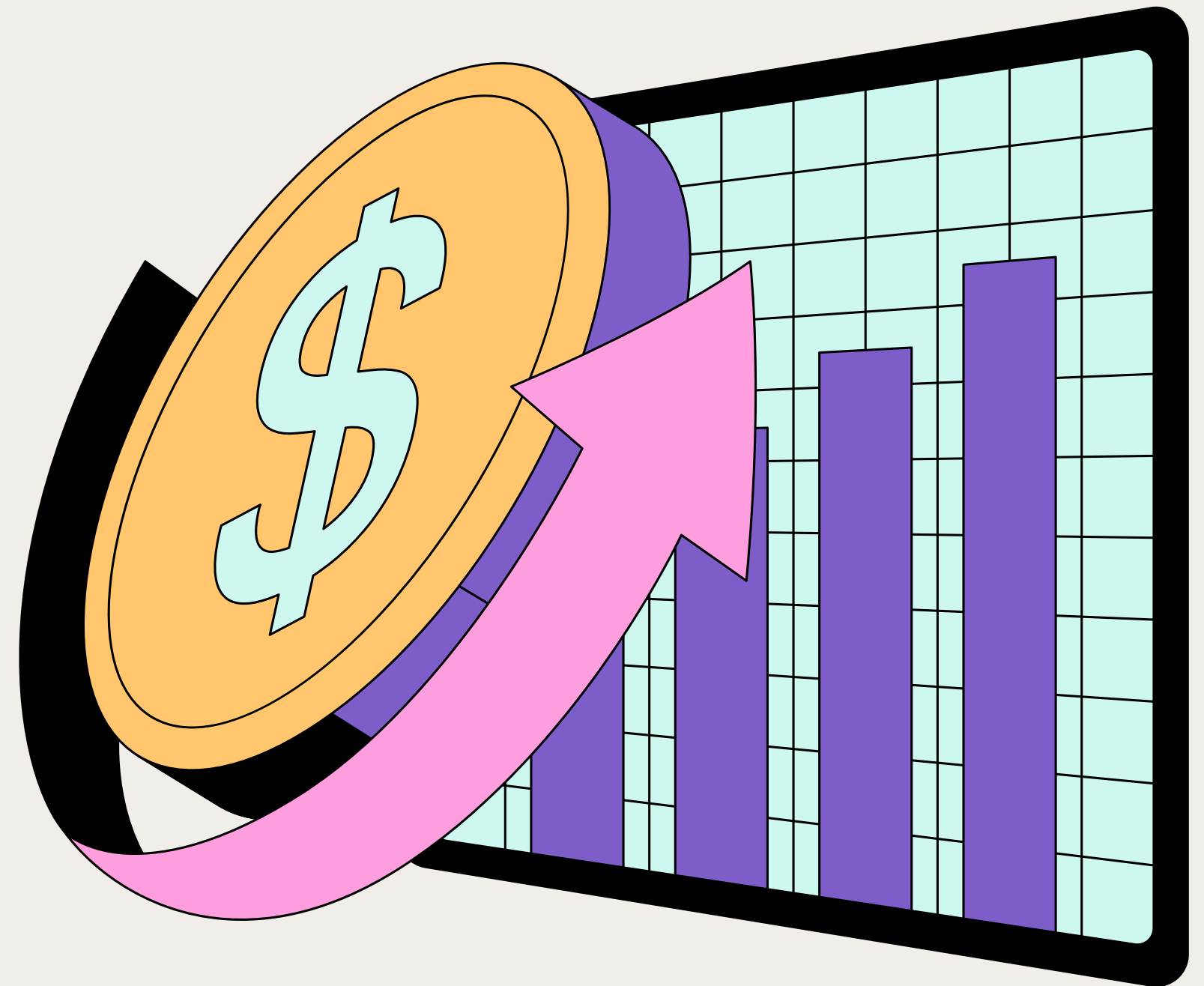


Super Store

Optimalisasi Profit dengan Machine Learning

Marcella Aurelia Yatijan - 71477



Tujuan

Memprediksi Profit dan Rugi
Order Berdasarkan Pola
Historis untuk Optimasi
Keputusan Bisnis

Mengetahui variabel yang
paling berpengaruh
terhadap profit



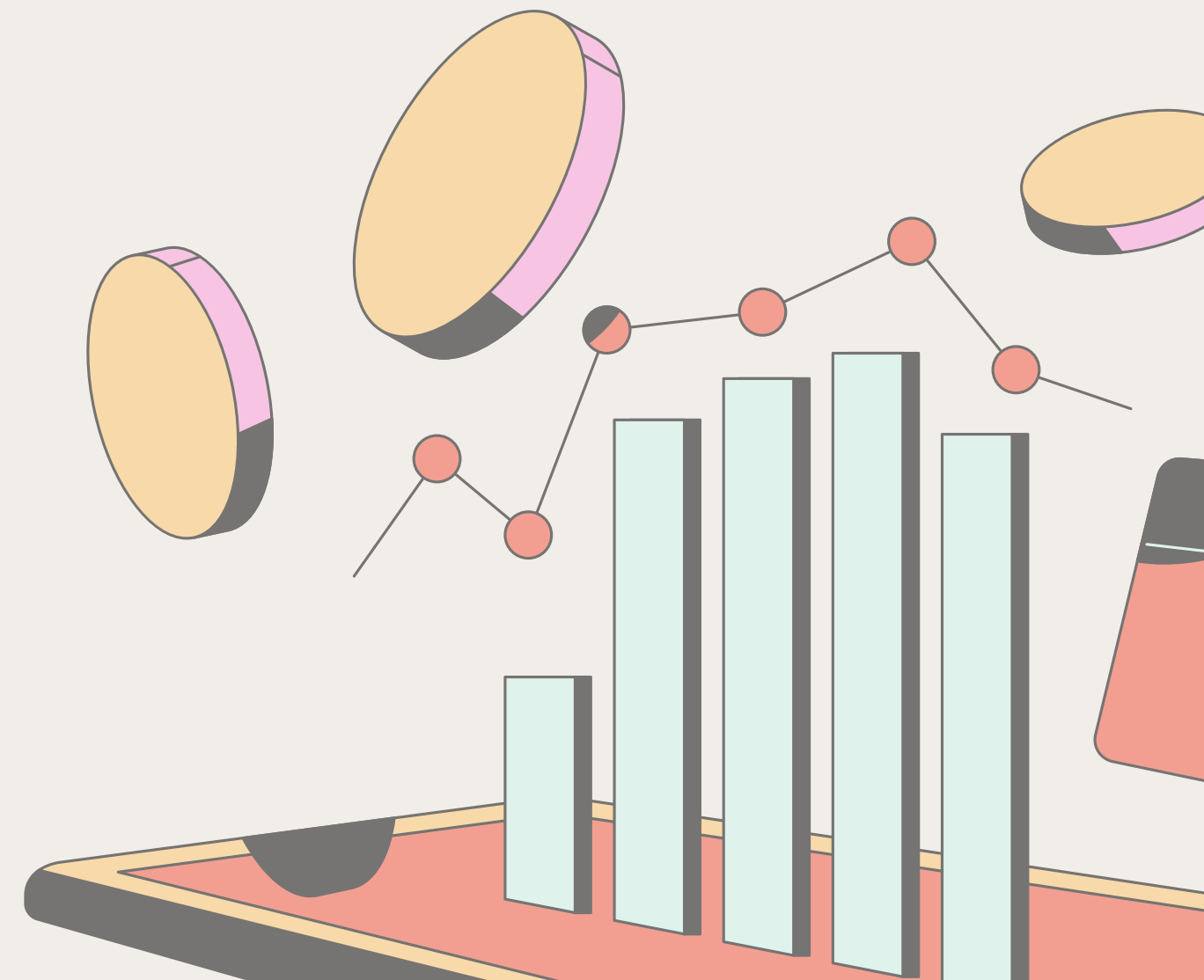
Dataset

FEATURE

- DISCOUNT
- DISCOUNT_BIN
- SALES
- CATEGORY
- MARKET
- SUB_CATEGORY
- ORDER_PRIORITY
- SEGMENT
- SHIPPING_COST
- PRODUCT_NAME
- COUNTRY
- REGION
- QUANTITY
- SHIP_DATE
- CUSTOMER_NAME
- SHIP_MODE

TARGET

- PROFIT_BIN



Profit dan Discount Bin

Kelas	Syarat	Penjelasan
'Rugi'	$p < 0$	Jika profit negatif (rugi)
'Untung Kecil'	$0 \leq p \leq 30$	Jika profit sedikit
'Untung Besar'	$p > 30$	Jika profit besar

=== Statistik Profit ===

Mean : 28.64

Median : 9.24

Modus : [0.0]

discount_bin	Range Diskon (discount)	Penjelasan
0	0.00	Tanpa diskon
1	>0.00 s.d. 0.10	Diskon rendah (0–10%)
2	>0.10 s.d. 0.30	Diskon sedang (10–30%)
3	>0.30 s.d. max_val (mungkin 0.8–1.0)	Diskon tinggi (>30%)



Feature Importance by Chi Squared

===== Hasil Uji Chi-Squared terhadap 'profit_bin' =====					
Feature	Chi2 Hitung	Chi2 Tabel	p-value	Kesimpulan	
discount_bin	36065.6884	12.5916	0.0000	✓	Tolak H0
product_name	28422.4752	7777.5746	0.0000	✓	Tolak H0
country	24149.4983	332.8538	0.0000	✓	Tolak H0
sub_category	10604.9665	46.1943	0.0000	✓	Tolak H0
category	6220.4128	9.4877	0.0000	✓	Tolak H0
ship_date	4812.2120	3052.9553	0.0000	✓	Tolak H0
region	3323.9449	36.4150	0.0000	✓	Tolak H0
customer_name	2643.4157	1681.8204	0.0000	✓	Tolak H0
market	1397.6546	21.0261	0.0000	✓	Tolak H0
order_priority	4.2271	12.5916	0.6460	Gagal	Tolak H0
ship_mode	3.1470	12.5916	0.7902	Gagal	Tolak H0
segment	1.0461	9.4877	0.9027	Gagal	Tolak H0



FEATURE

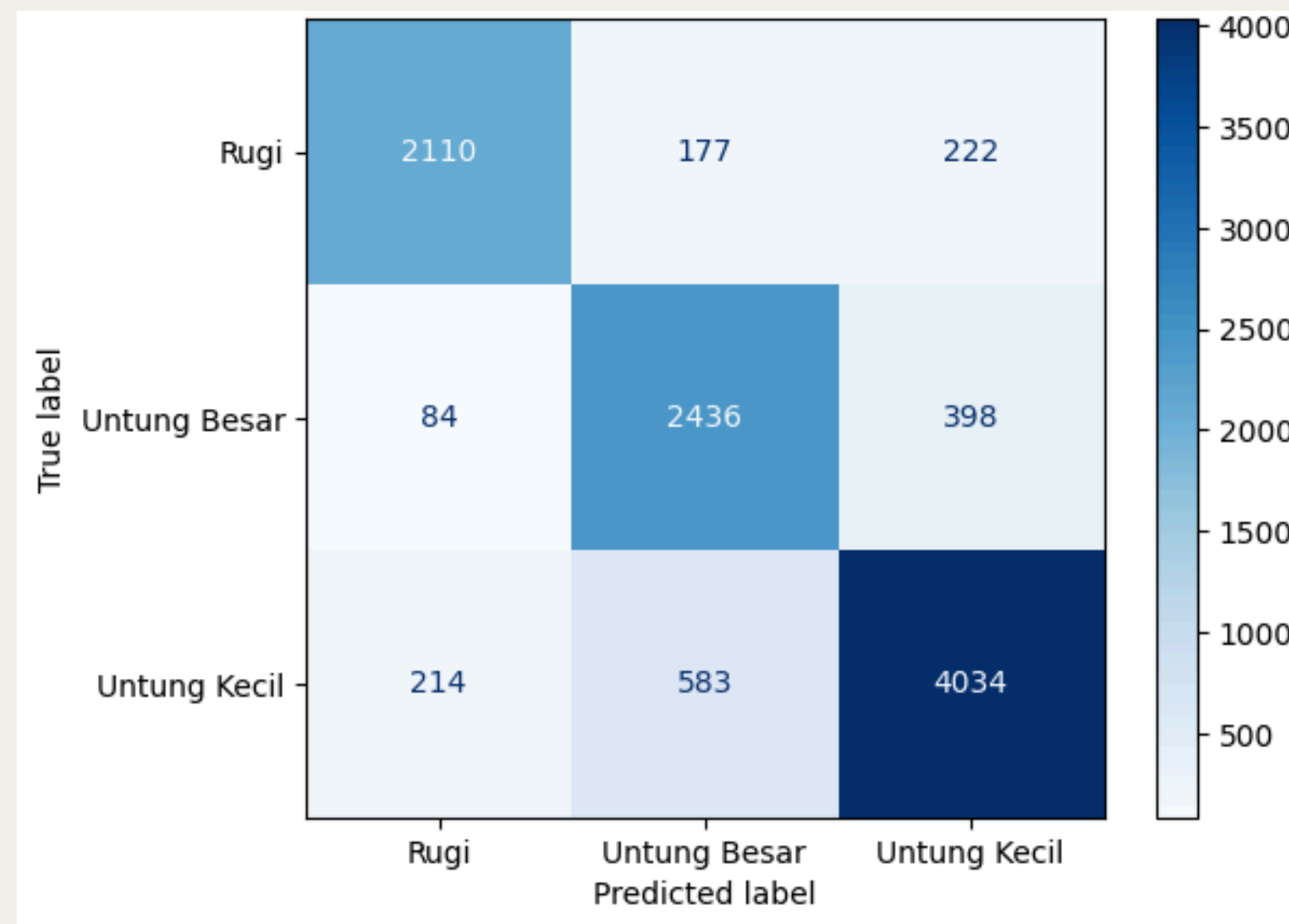
- DISCOUNT
- SALES
- CATEGORY
- MARKET
- SUB_CATEGORY
- SHIPPING_COST
- PRODUCT_NAME
- COUNTRY
- REGION
- QUANTITY
- SHIP_DATE
- CUSTOMER_NAME

Model Accuracy

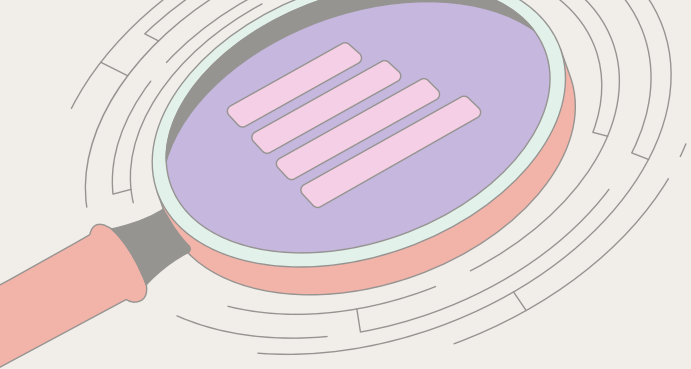
Model	Accuracy
Random Forest	84%
XGBoost + Optuna	84%
XGBoost	83%
StackingClassifier	83.4%
BaggingClassifier	82%
GradientBoostingClassifier	82%
VotingClassifier	77%
KNN	58%



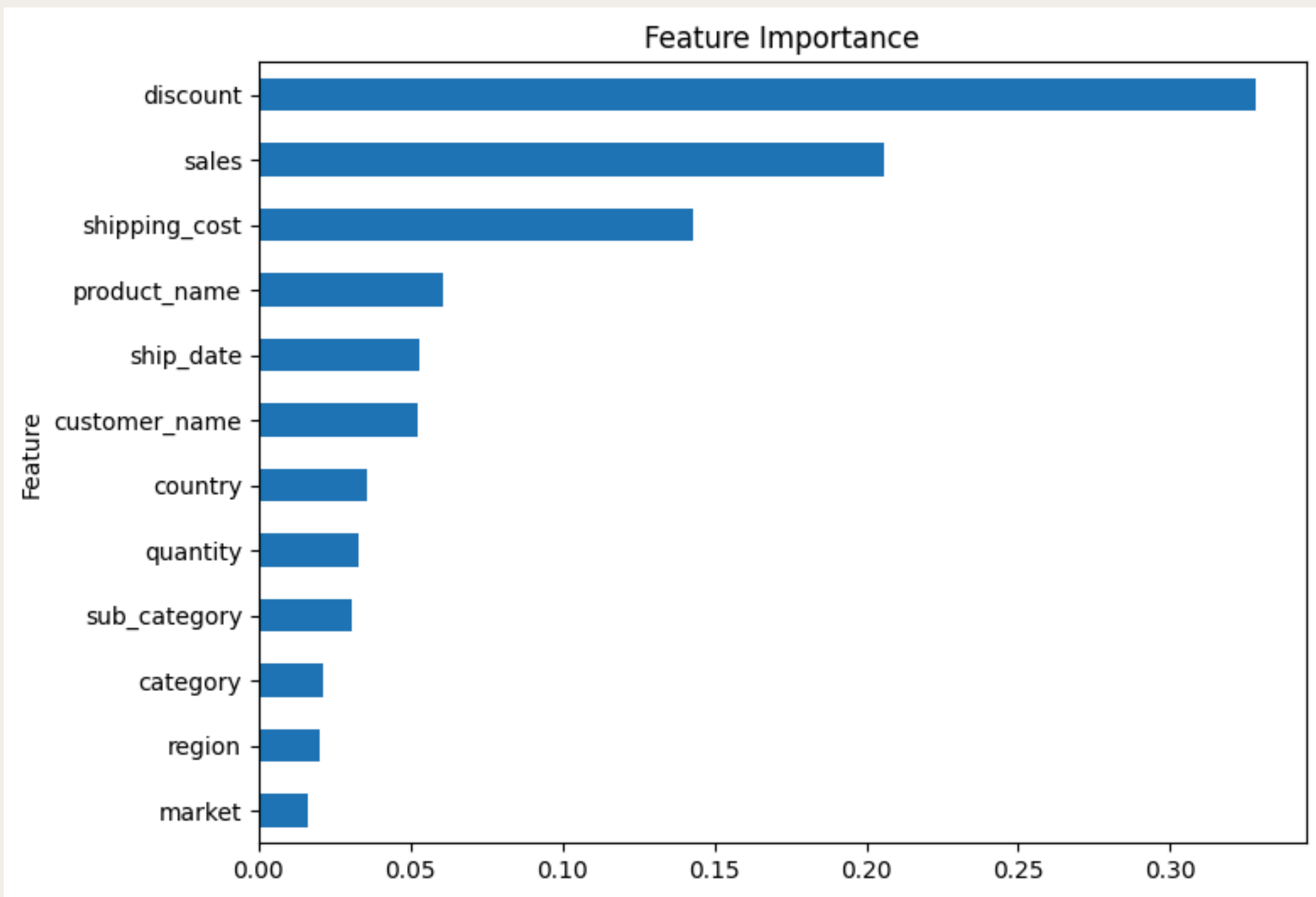
Random Forest



ACCURACY 84%
ROC AUC SCORE (OVR) 0.9562



Feature Importance by Random Forest

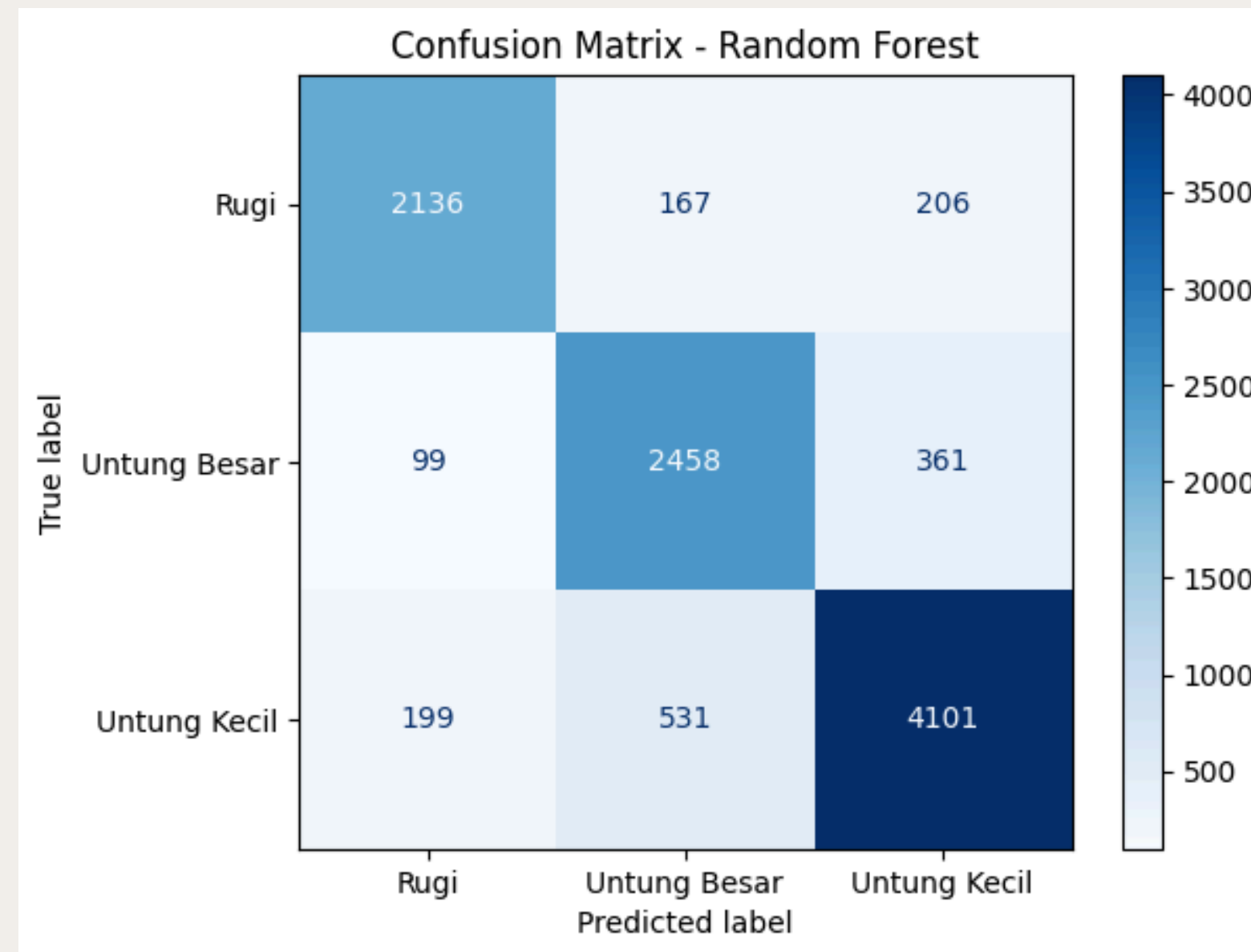


Feature	Chi2 Hitung
discount_bin	36065.6884
product_name	28422.4752
country	24149.4983
sub_category	10604.9665
category	6220.4128
ship_date	4812.2120
region	3323.9449
customer_name	2643.4157
market	1397.6546
order_priority	4.2271
ship_mode	3.1470
segment	1.0461

Model dengan 6 Feature

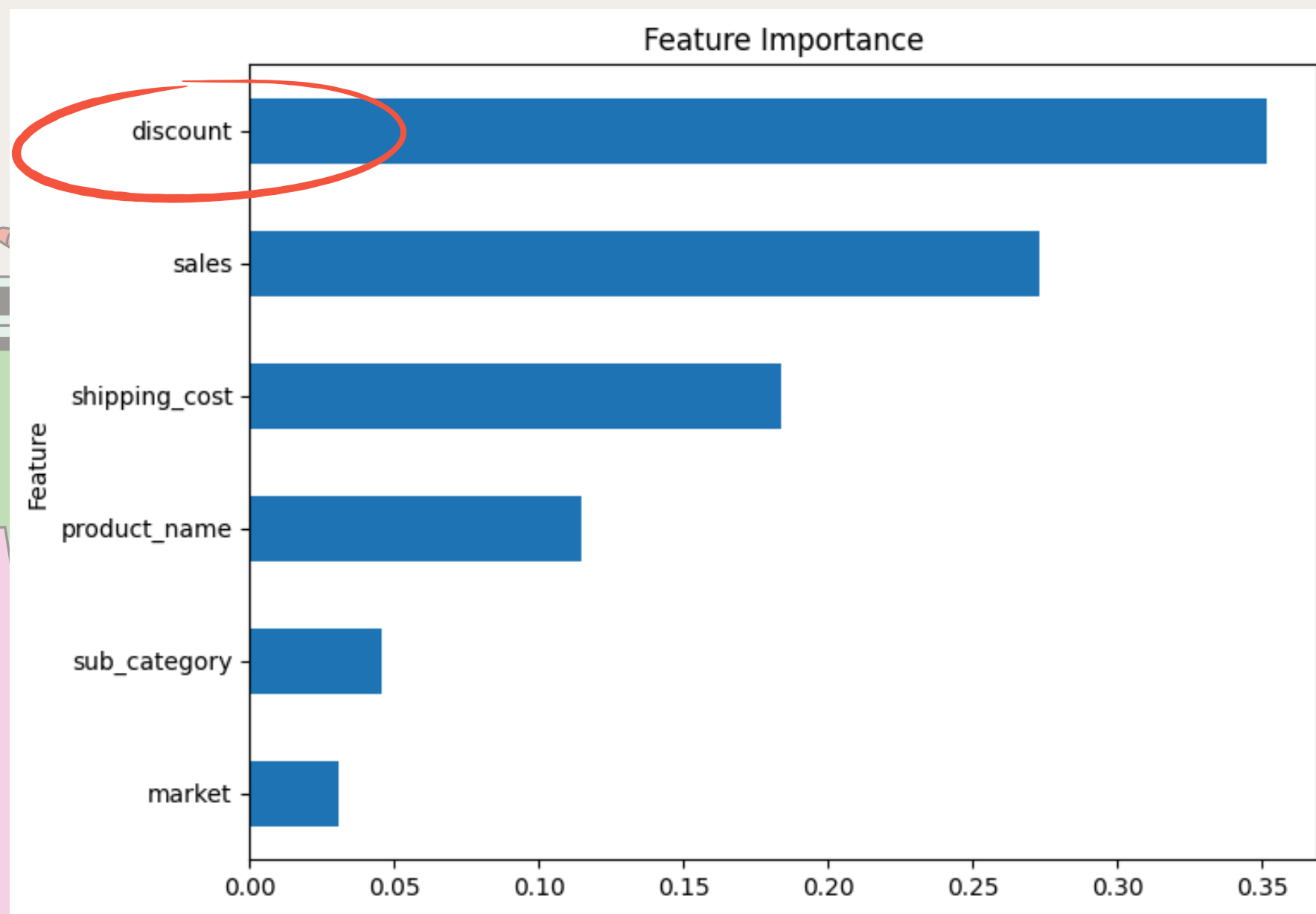
FEATURE

- DISCOUNT
- SALES
- MARKET
- SUB_CATEGORY
- SHIPPING_COST
- PRODUCT_NAME
- ~~COUNTRY~~
- ~~REGION~~
- ~~QUANTITY~~
- ~~SHIP_DATE~~
- ~~CUSTOMER_NAME~~
- ~~CATEGORY~~



ACCURACY 85%
ROC AUC SCORE (OVR) 0.9616

Model dengan 6 Feature



```
=== Feature Importance ===  
Feature RF Importance  
0 discount 0.351595  
1 sales 0.272951  
2 shipping_cost 0.184113  
3 product_name 0.115046  
4 sub_category 0.045771  
5 market 0.030524
```

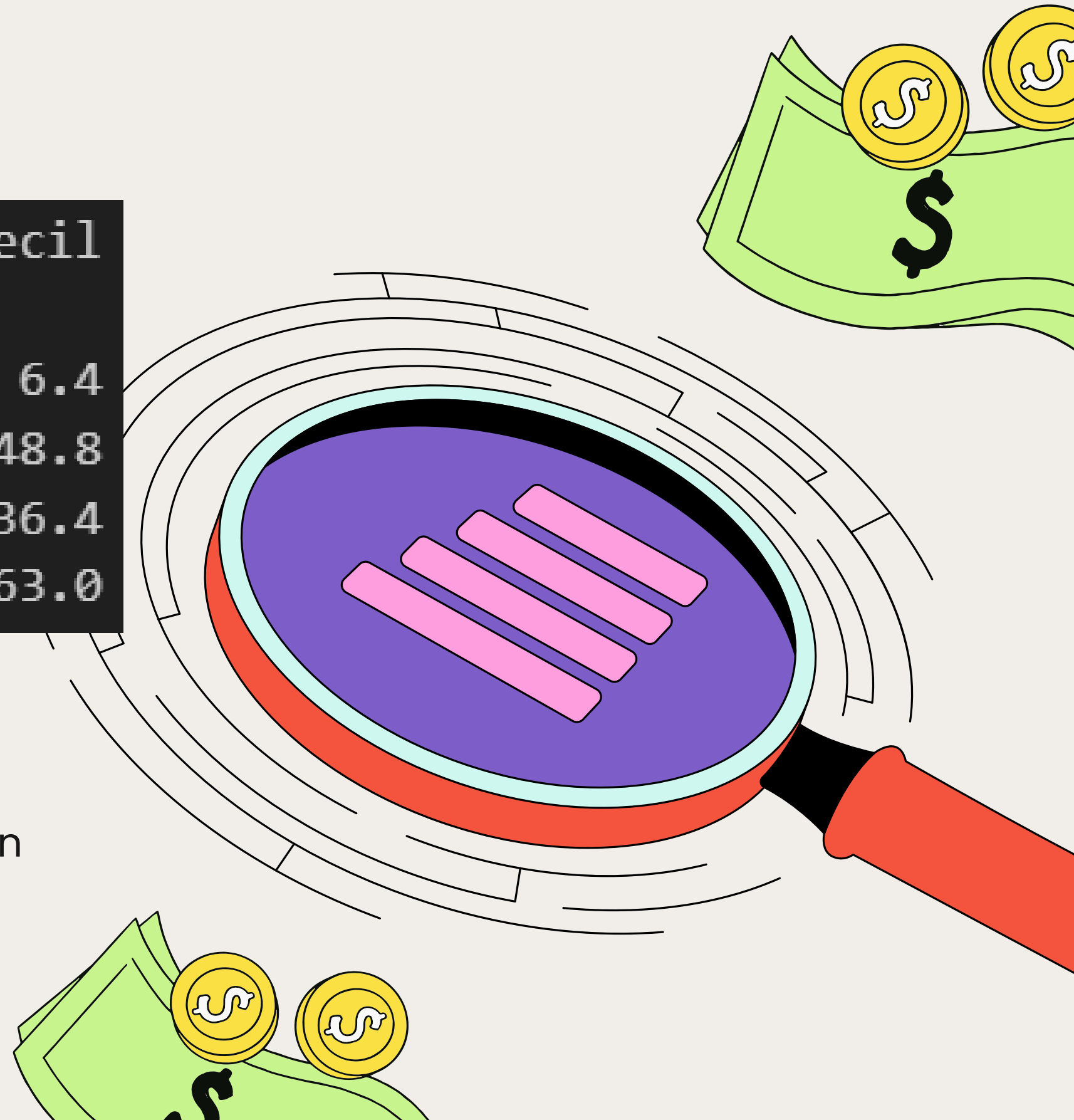
Diskon masih menjadi variabel utama yang mempengaruhi profit

Analisis Lanjutan

profit_bin	Rugi	Untung Besar	Untung Kecil
discount_bin			
3	92.5	1.2	6.4
2	28.5	22.7	48.8
1	19.2	44.4	36.4
0	0.0	37.0	63.0

92.5% transaksi dengan diskon di atas 30% berakhir **Rugi!**
Hanya 1.2% yang Untung Besar, dan 6.4% Untung Kecil.

✓ Kesimpulan: Diskon terlalu tinggi justru menyebabkan kerugian besar. Ini sinyal bahaya untuk strategi pricing.



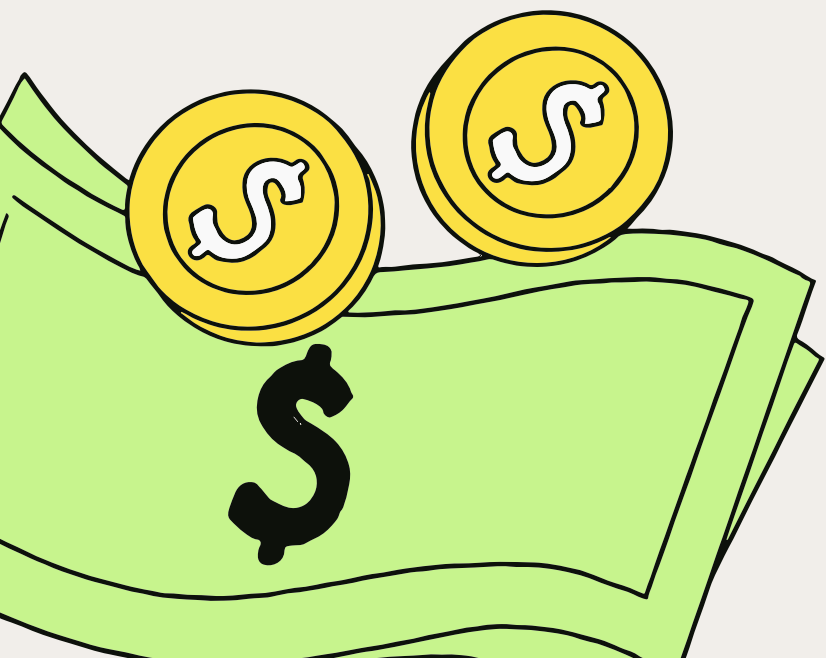
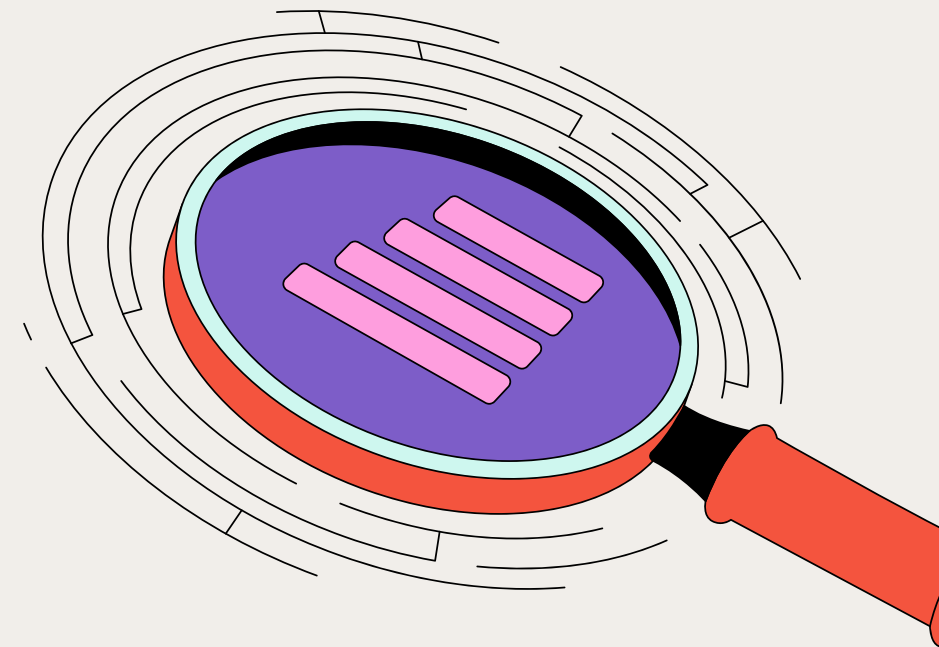


PERLU PENDEKATAN BARU

Optimalisasi Besar
Diskon

**TANPA MENGORBANKAN
MARGIN**

Maksimalkan diskon
kurang dari 30%



Kesimpulan

Accuracy model prediksi profit 85%

Diskon tinggi → justru meningkatkan beban rugi.

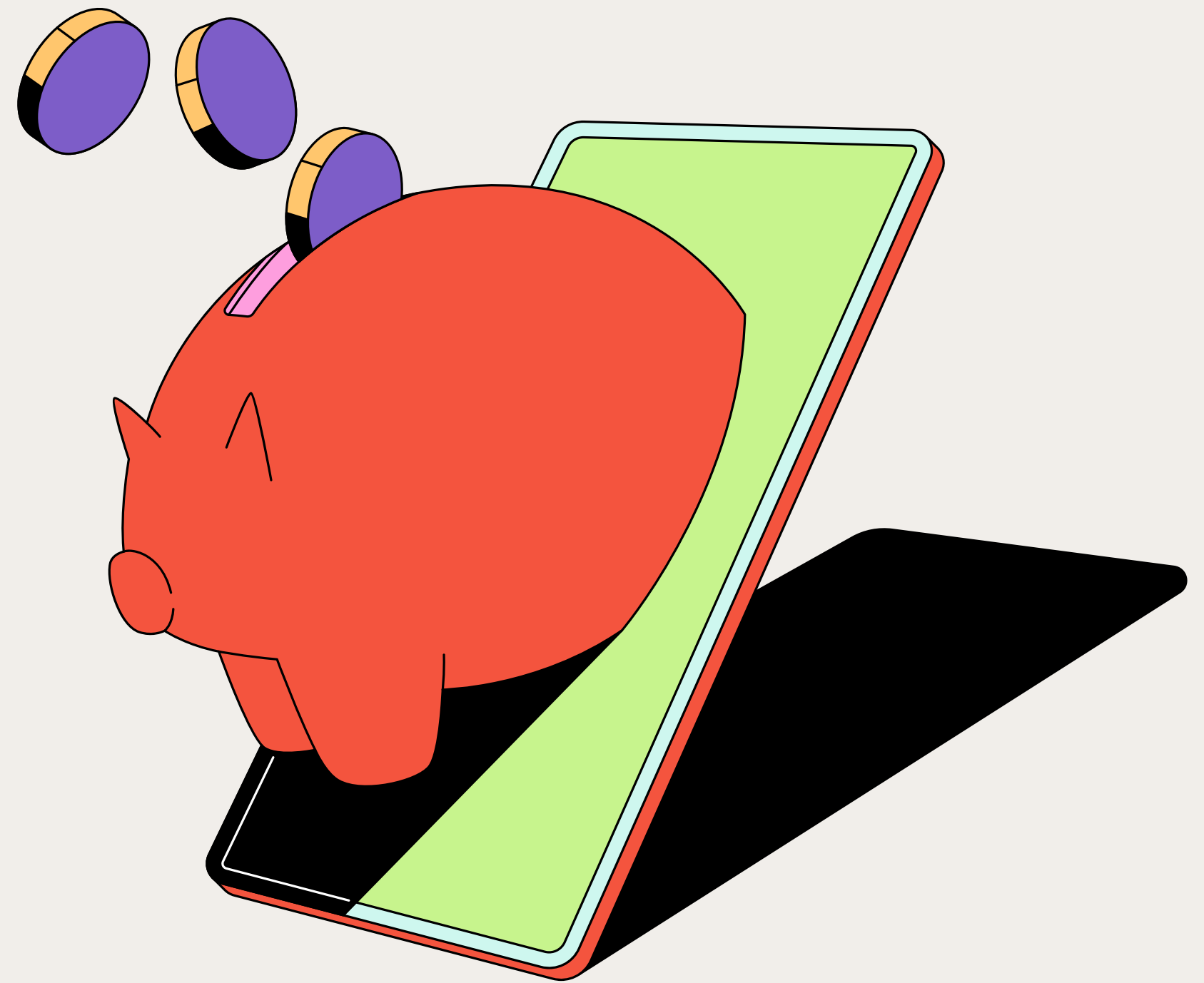
Evaluasi kembali pengaruh diskon terhadap sales

By Data, diskon kurang dari 30% menghasilkan profit yang cukup besar



Ide Tambahan

(Perlu Analisis Lebih Lanjut)



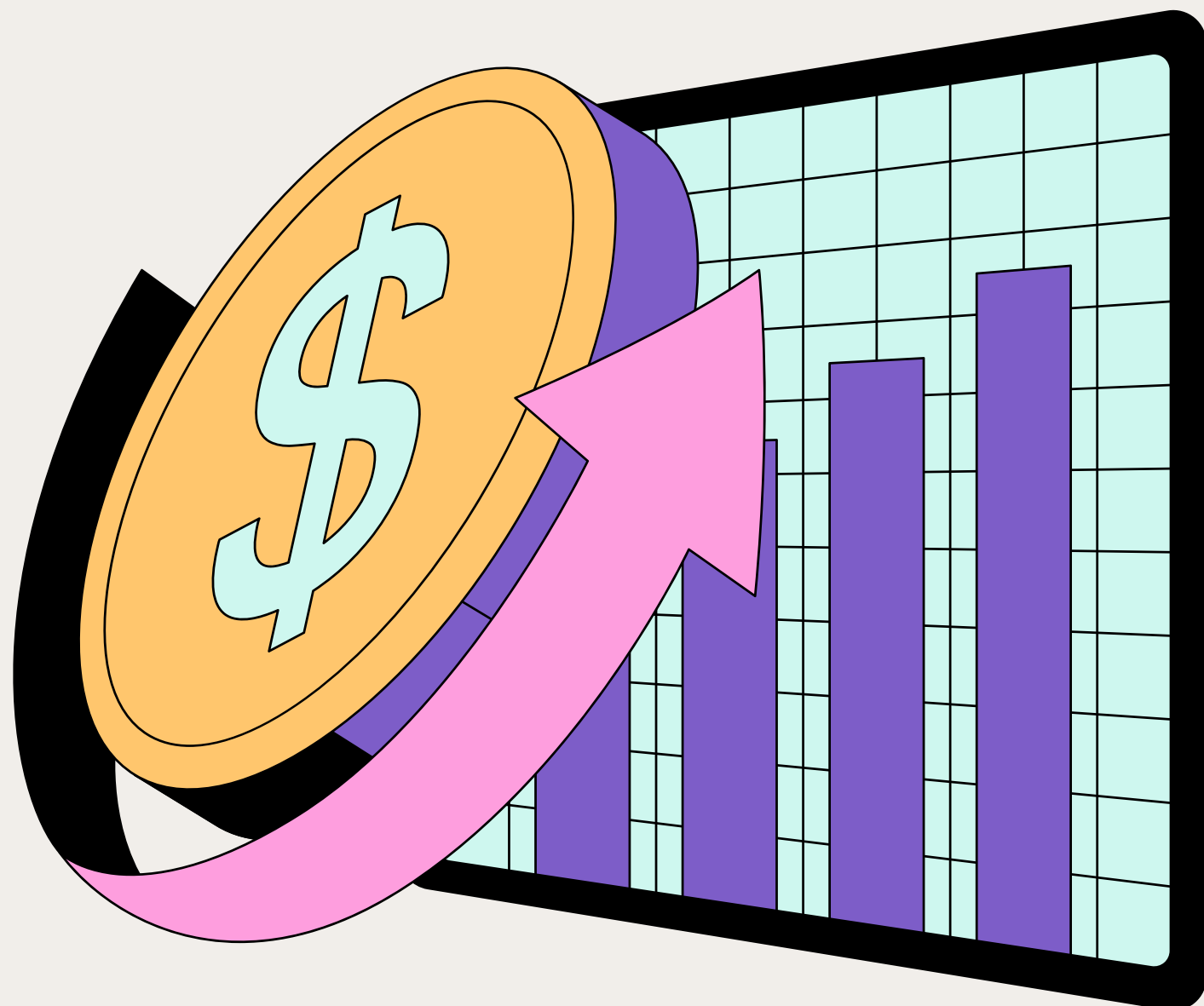
Penerapan SLA (Service Level Agreement) untuk pengiriman



Customer akan mendapat voucher discount hanya apabila mengalami keterlambatan pengiriman

Discount diberikan secara selektif sehingga dapat mengurangi budget discount di awal yang besar





Benefit



Cost Effective

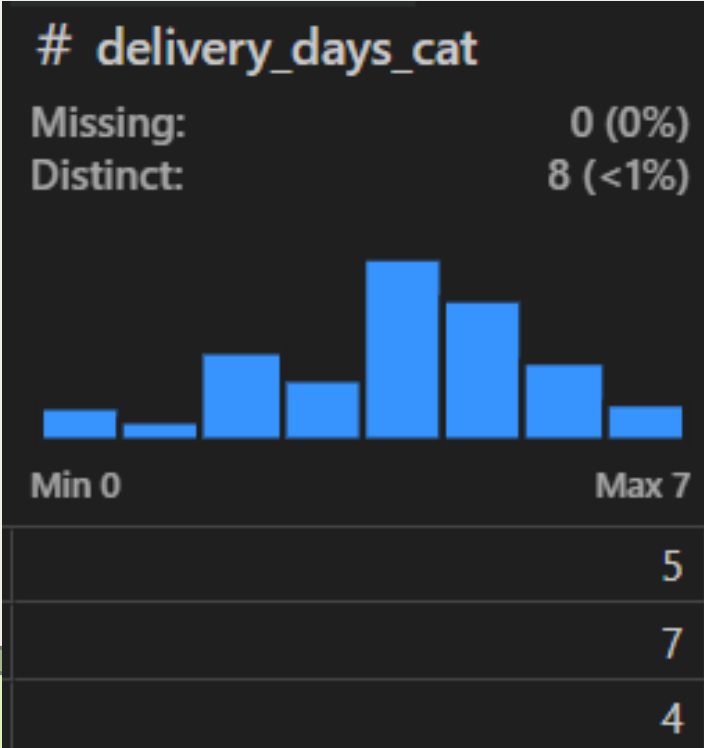


Menciptakan Repeat Order (Perlu pembuktian analisis lebih lanjut)



Meningkatkan Loyalitas Pelanggan (Perlu pembuktian analisis lebih lanjut)

Membangun Model Delivery Speed



delivery_days_cat	
4 hari	14434
5 hari	11221
2 hari	7026
6 hari	6255
3 hari	5035
7 hari	3057
0 hari	2600
1 hari	1662

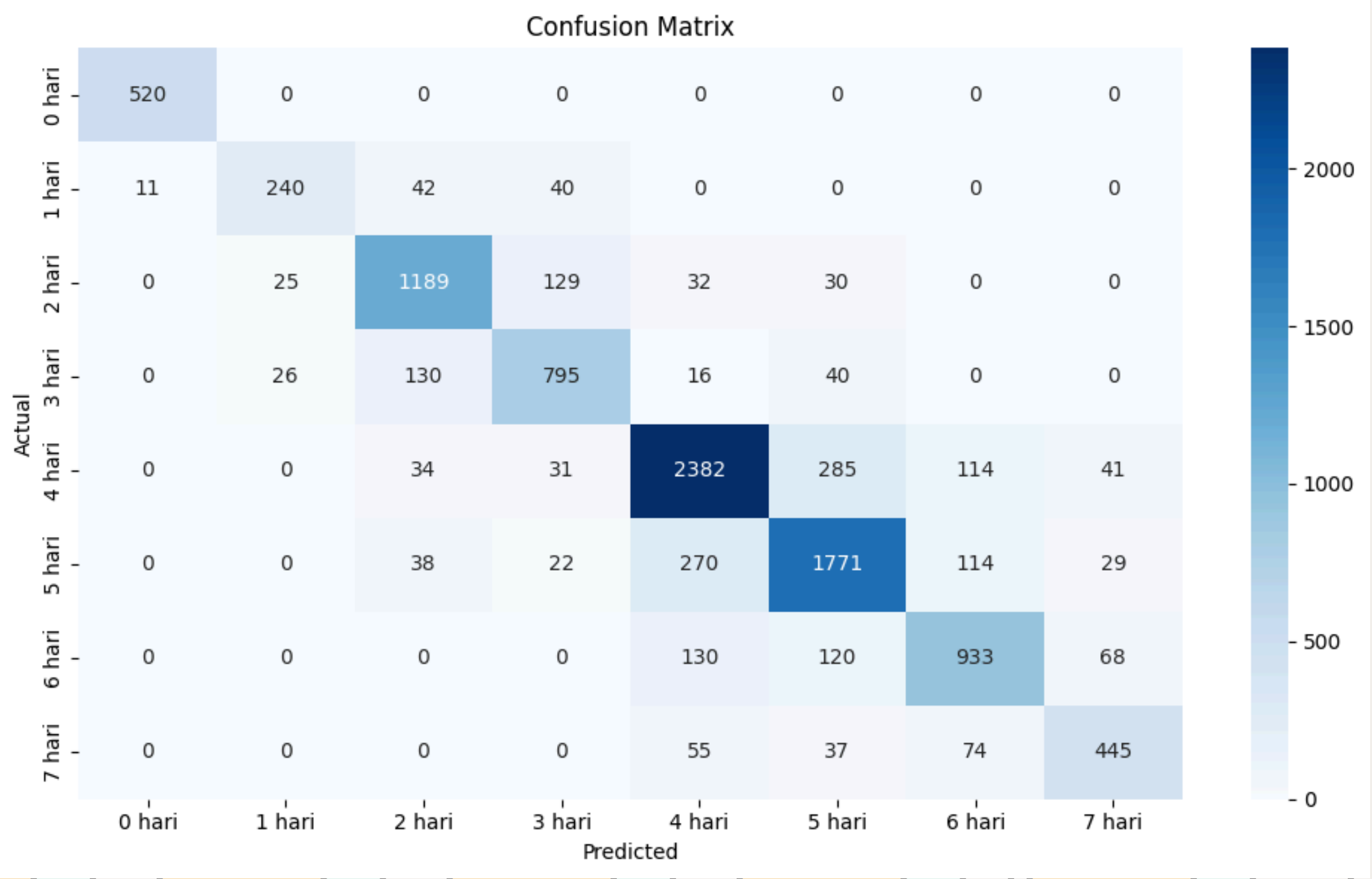
Chi Squared Test Of Independence

===== Hasil Uji Chi-Squared terhadap 'Delivery Speed' =====					
Feature	Chi2 Hitung	Chi2 Tabel	p-value	Kesimpulan	
segment	44.9272	23.6848	0.0000	✓	Tolak H0
region	319.3392	106.3948	0.0000	✓	Tolak H0
ship_mode	93450.7180	32.6706	0.0000	✓	Tolak H0
order_priority	30136.6393	32.6706	0.0000	✓	Tolak H0
country	2798.0755	1097.4844	0.0000	✓	Tolak H0
ship_date	38617.3511	10477.5353	0.0000	✓	Tolak H0
market	153.1353	58.1240	0.0000	✓	Tolak H0
customer_name	16127.6794	5732.5505	0.0000	✓	Tolak H0
order_id	350378.3895	176212.9057	0.0000	✓	Tolak H0
category	22.2720	23.6848	0.0732	Gagal	Tolak H0
product_name	26720.1248	26888.8717	0.1795	Gagal	Tolak H0
sub_category	122.4037	137.7015	0.2361	Gagal	Tolak H0

Model Prediction

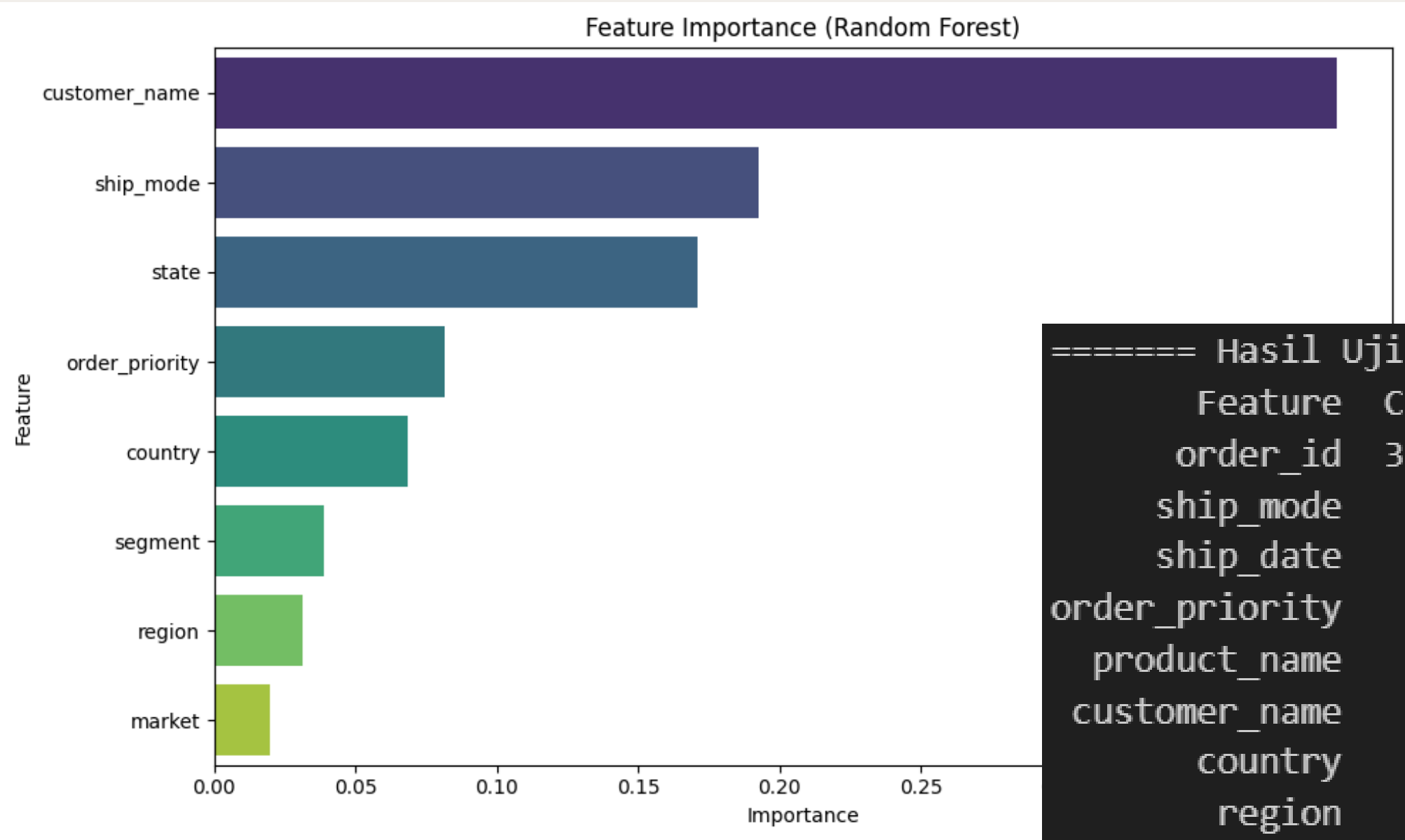
Accuracy: 81%

ROC AUC Score: 0.9665



Feature Importance

Accuracy: 81%



===== Hasil Uji Chi-Squared terhadap 'Delivery Speed' =====					
Feature	Chi2 Hitung	Chi2 Tabel	p-value	Kesimpulan	
order_id	350378.3895	176212.9057	0.0000	Tolak	H0
ship_mode	93450.7180	32.6706	0.0000	Tolak	H0
ship_date	38617.3511	10477.5353	0.0000	Tolak	H0
order_priority	30136.6393	32.6706	0.0000	Tolak	H0
product_name	26720.1248	26888.8717	0.1795	Gagal	Tolak H0
customer_name	16127.6794	5732.5505	0.0000	Tolak	H0
country	2798.0755	1097.4844	0.0000	Tolak	H0
region	319.3392	106.3948	0.0000	Tolak	H0
market	153.1353	58.1240	0.0000	Tolak	H0
sub_category	122.4037	137.7015	0.2361	Gagal	Tolak H0
segment	44.9272	23.6848	0.0000	Tolak	H0
category	22.2720	23.6848	0.0732	Gagal	Tolak H0

Kesimpulan

Accuracy model prediksi profit 85%

Accuracy model prediksi delivery speed 81%

Diskon tinggi $\neq$ loyalitas $\rightarrow$ justru meningkatkan beban rugi.

Penerapan voucher berbasis SLA memberikan win-win solution: biaya lebih terkendali, pelanggan tetap puas.



Action Plan

Evaluasi Kembali Diskon dan Model Machine Learning

PIC	Langkah	Deskripsi
Data/IT	Integrasi model prediksi profit & delivery	Gabungkan ke sistem operasional/sales pipeline
Marketing	Uji coba strategi SLA-voucher	Terapkan ke sebagian pelanggan high-priority
Finance & CS	Monitoring hasil	Evaluasi dampak ke margin dan retensi pelanggan
Management	Roll-out penuh	Ekspansi strategi ke semua area



Let's

Work Together

Thank You For Your Attention

